



DAM

WEB INTERFACE DESIGN

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BOOK



Diseño de interfaces web

2.ª edición
revisada y actualizada

MATERIALS

GitHub: <https://github.com/BegoRodriguez/DAW-Web-Interfaces-Design>

Dropbox:

https://www.dropbox.com/scl/fo/kixyrup7y75gxj8g2fa8i/AHFHf_YxbQpdpjtZ8WfCR2Y?rlkey=fplc4rmte84t7003t10qcdjto&dl=0

- Link to contents of DAW - Web Interfaces Design:
https://www.dropbox.com/scl/fo/kixyrup7y75gxj8g2fa8i/AHFHf_YxbQpdpjtZ8WfCR2Y?rlkey=fplc4rmte84t7003t10qcdjto&dl=0

Pinned

Customize your pins

 **DAM-Sistemas-informaticos** Public

Materials para el módulo Sistemas Informáticos del IES Ribera de Castilla de Primero de DAM

 **SMR-Innovacion** Public

Materials para el módulo Innovación aplicada al sector productivo del IES Ribera de Castilla de Primero de SMR

 **DAW-Web-Interfaces-Design** Public

Materials for Web Interfaces Design in IES Ribera de Castilla

 **Recursos** Public

Compilación de recursos gratuitos de programación, bases de datos y tecnología en general.

 8  3

TEAMS

I have created a classroom team for communication and project and activities delivery on Teams.

This is the code:

|4s6zs6

USER INTERFACE DESIGN FOR WEB DEVELOPMENT

Goals:

- Recognize the importance of visual communication and its basic principles.
- Analyze different alternatives for presenting information in web documents.
- Identify and analyze the elements involved in creating web prototypes.
- Use and analyze content management systems as the foundation for developing a website.
- Plan and establish the process for designing a website.

DESIGN PRINCIPLES BASED ON HUMAN-COMPUTER INTERACTION

Human-Computer Interaction (HCI) is a field of study that focuses on the design and development of interactive systems that are easy and efficient to use.

HCI guidelines are recommendations that can be used to design and develop user interfaces that are more user-friendly and effective.

LICKLIDER & CLARK (1962)

Ten **problems to solve** for better Human–Computer Interaction (HCI):

- Sharing computer time among many users.
- An input–output system for communication using symbolic and graphical data.
- An interactive system capable of processing operations in real time.
- Systems for massive information storage that allow rapid retrieval.
- Systems that facilitate cooperation among people in the design and programming of large systems.

LICKLIDER & CLARK (1962) (II)

Ten **problems to solve** for better Human-Computer Interaction (HCI) (II):

- Recognition by computers of voice, printed handwriting, and direct handwritten data input.
- Automatic translation between natural languages.
- Automatic programming — enabling computers to generate or assist in writing programs.
- Self-organizing systems capable of adapting and improving their own performance.
- Logical and heuristic problem-solving systems that support human reasoning.

LICKLIDER & CLARK (1962) (III)

These ten problems anticipated almost everything modern computing tries to solve:

- multitasking
- graphical interfaces
- real-time systems
- databases
- collaborative development
- speech recognition
- machine translation
- AI-assisted coding
- adaptive systems
- expert systems

HANSEN (1972)

Wilfred J. Hansen's four principles for user-interface design, proposed in 1971, they are an important early contribution to Human-Computer Interaction (HCI).

Hansen presented them in “User Engineering Principles for Interactive Systems”, published in the proceedings of the 1971 Fall Joint Computer Conference.

The four principles are:

1. Know the user
2. Minimize memorization
3. Optimize operations
4. Engineer for errors

1. KNOW THE USER

The interface should be designed around the people who will actually use it, rather than around what is technically convenient for the programmer.

The designer should understand:

- the user's goals and needs;
- their previous experience;
- their knowledge and skills;
- the context in which they work;
- their limitations and likely difficulties.

2. MINIMIZE MEMORIZATION

Users should not have to remember unnecessary information in order to operate the system.

The interface should provide information that the user needs at the moment they need it.

For example, instead of requiring:

"Remember that Ctrl+Alt+N creates a new network."

the system could provide a visible New network button or menu item.

This principle is closely related to a major modern HCI idea: recognition is generally easier than recall.

2. MINIMIZE MEMORIZATION (II)

You can see its influence in things we now take for granted:

- menus;
- visible buttons;
- autocomplete;
- contextual help;
- suggestions;
- recently used items;
- meaningful icons and labels.

3. OPTIMIZE OPERATIONS

The system should make users' actual tasks efficient.

This means reducing unnecessary actions, avoiding cumbersome procedures and designing operations around the user's workflow.

Design the system so that the user's work can be accomplished efficiently and naturally.

This is particularly relevant today for usability, keyboard shortcuts, automation, defaults and reducing the number of steps required to accomplish frequent tasks.

4. ENGINEER FOR ERRORS

Hansen recognized that users will make mistakes, and that a good system should therefore be designed to anticipate and handle them.

The important point is that errors should not simply be treated as the user's fault. For example, if a user enters: 192.168.1.300
a good interface should detect the invalid address and explain the problem.

An even better interface might prevent invalid input from being entered in the first place.

4. ENGINEER FOR ERRORS (II)

This principle leads to several modern practices:

- validation.
- meaningful error messages.
- confirmation for dangerous operations.
- undo.
- recovery mechanisms.
- preventing invalid actions.
- preserving user data after an error.

CORE FUNCTIONS OF GRAPHIC DESIGN

Graphic design isn't just "making things pretty." In web development, it has functional, communicative, and cognitive purposes.

- **Communication & Information Delivery:** Translates raw data, copy, or abstract concepts into structured, easily digestible visual messages (think infographics, road signs, or annual reports).
- **Persuasion & Marketing:** Drives action and influences audience behavior in advertising, campaigns, and product packaging by appealing to specific emotions and needs.
- **Identity & Brand Building:** Establishes the distinct visual personality of a company, product, or organization through consistent logos, typography, color schemes, and imagery.

CORE FUNCTIONS OF GRAPHIC DESIGN (II)

Graphic design isn't just "making things pretty." In web development, it has functional, communicative, and cognitive purposes.

- **User Experience (UX) & Navigation:** Guides people through digital spaces (websites, apps) and physical environments (wayfinding in airports or subways) by creating intuitive visual hierarchies.
- **Engagement & Aesthetics:** Captures attention in a visual-heavy world, making content memorable and pleasant to interact with.

GROUPS OF ELEMENTS

- **Conceptual elements:** The ideas, concepts and meanings that the interface represents. Example: the concept of "deleting a file."
- **Visual elements:** What the user actually perceives: colours, typography, icons, shapes, images, etc. Example: a trash-bin icon representing deletion.
- **Relational elements:** The relationships between elements: hierarchy, proximity, alignment, grouping, sequence and navigation. Example: placing "Save" next to "Cancel", or grouping network settings together.
- **Practical elements:** The elements that allow the user to perform actions and accomplish tasks. Example: buttons, forms, sliders, menus and controls.

DEFINITION OF WEB INTERFACE

For all the above, we can conclude that:

*“A web interface is the set of **visual and interactive elements**, together with their **organization and layout**, that allows users to access information, navigate a website and interact with its content and functions efficiently.”*

If these elements are not carefully designed and coordinated, a website may still work technically, but achieving a good user experience will largely be a matter of chance.

EXAMPLES OF GOOD AND BAD DESIGN

DE JUEVES A SABADOS DE 20.30 A 00.00 HRS. **SABADOS Y**
DOMINGOS DE 12.30 A 14.30 HRS.

FUSILES AL FERRETTO

RABAS A LA PROVENZAL

CARACOLLES A LA BORDALEZA

RABAS A LA CALABRIA

BERLUZA AL AJILLO

GAMBAS AL AJILLO

ABIERTO DE JUEVES A SABADOS DE NOCHE- SABADOS Y DOMINGOS ALMUERZOS

HORARIO CENA DE 20.30 A 23.30 HRS.

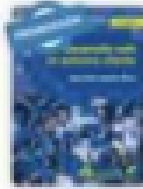
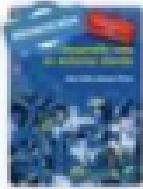
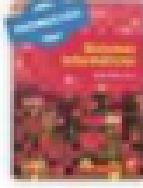
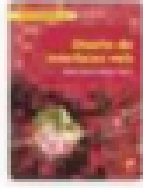
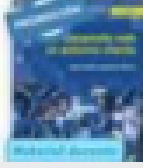
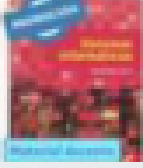
HORARIO ALMUERZO DE 12.30 A 14.30 HRS.



EDITORIAL SINTESIS

ENTRAR POR TITULO, AUTOR O ISBN

ORDENAR POR: **ULTIMOS LIBROS**

	Monica Pérez, Juan Carlos 9788410177008 Hasta 01/01/24 Desarrollo web en entorno cliente (2.ª edición) 142 páginas de 204		Monica Pérez, Juan Carlos 9788410177008 Hasta 01/01/24 Desarrollo web en entorno cliente (2.ª edición) 142 páginas de 204
	Rosa Arce, Jesús 9788410177015 Hasta 01/01/24 Sistemas Informáticos 42 páginas de 106		Gerardo Miguel López, Diana 9788410177015 Hasta 01/01/24 Diseño de interfaces web 11 páginas de 20
	Monica Pérez, Juan Carlos 9788410177008 Hasta 01/01/24 Desarrollo web en entorno cliente (2.ª edición) 142 páginas de 204		Rosa Arce, Jesús 9788410177015 Hasta 01/01/24 Sistemas Informáticos 42 páginas de 106

THE 6 PHASES OF WEBSITE DESIGN

The web design process moves from abstract strategy to concrete implementation in six distinct stages.

01. STRATEGY

Discovery, goals, & audience definition

02. STRUCTURE

Sitemap, user flows, & wireframes

03. UI DESIGN

Visual style, design system, & mockups

04. BUILD

Front-end, back-end, & CMS integration

05. TEST

QA, performance, accessibility, & cross-browser

06. LAUNCH

Deployment & continuous iteration

PHASE 1: DISCOVERY & STRATEGY

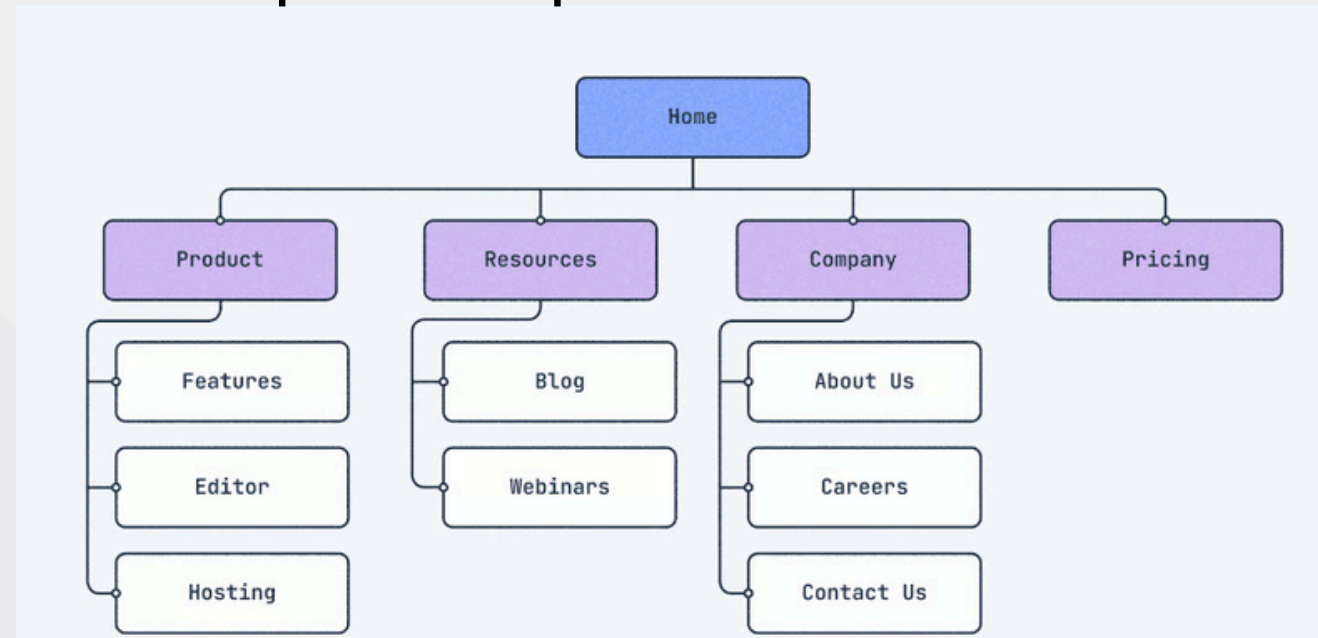
- **Objective:** Define the project scope, business goals, and target audience.
- **Key Tasks:**
 - Conduct stakeholder interviews
 - **Analyze competitors**
 - **Create user personas**
 - Establish Key Performance Indicators (KPIs).

PHASE 2: ARCHITECTURE & WIREFRAMING (UX)

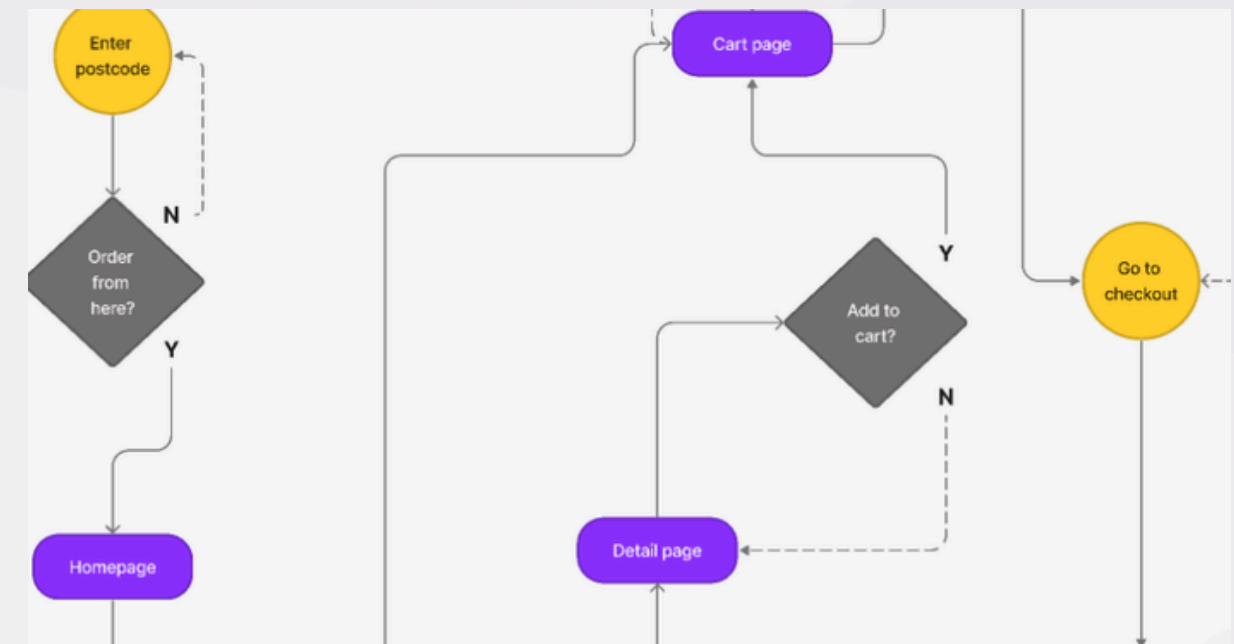
- **Objective:** Map out how content is organized and how users navigate the site.
- **Key Deliverables:**
 - **Sitemap:** A structural tree of all pages.
 - **User Flows:** Step-by-step paths users take to complete key tasks (e.g., checkout, signup).
 - **Wireframes:** Low-fidelity grayscale blueprints focusing on element placement without visual styling.

PHASE 2: ARCHITECTURE & WIREFRAMING (UX) (II)

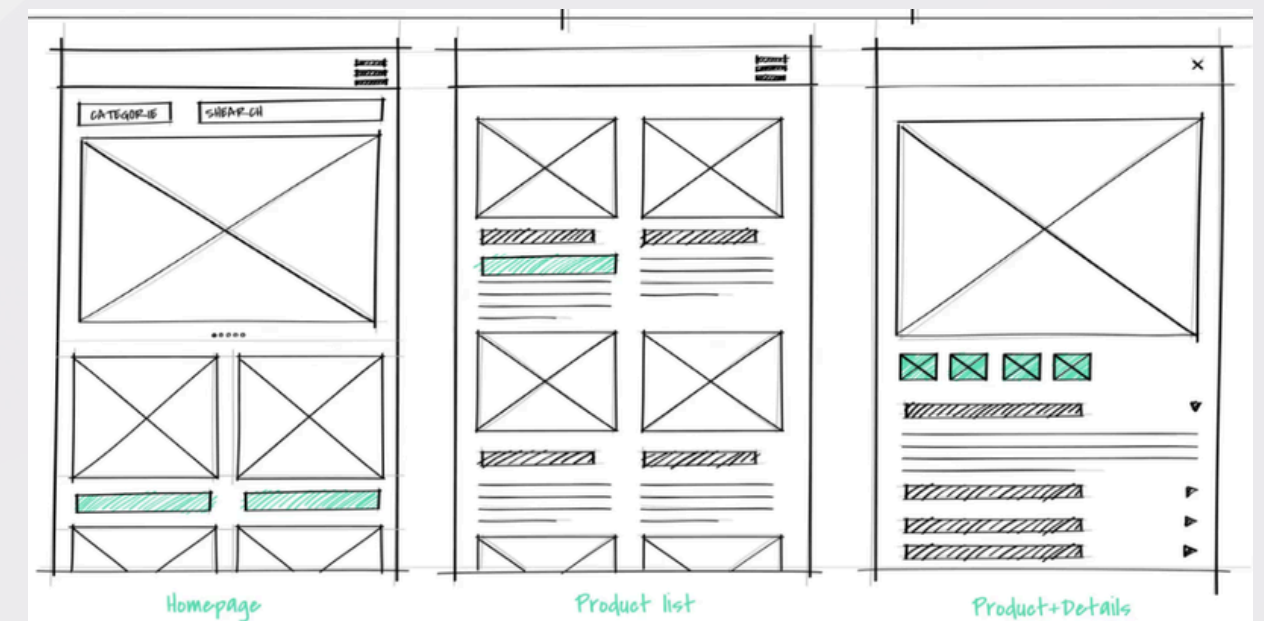
Sitemap example



User flow



Wireframe

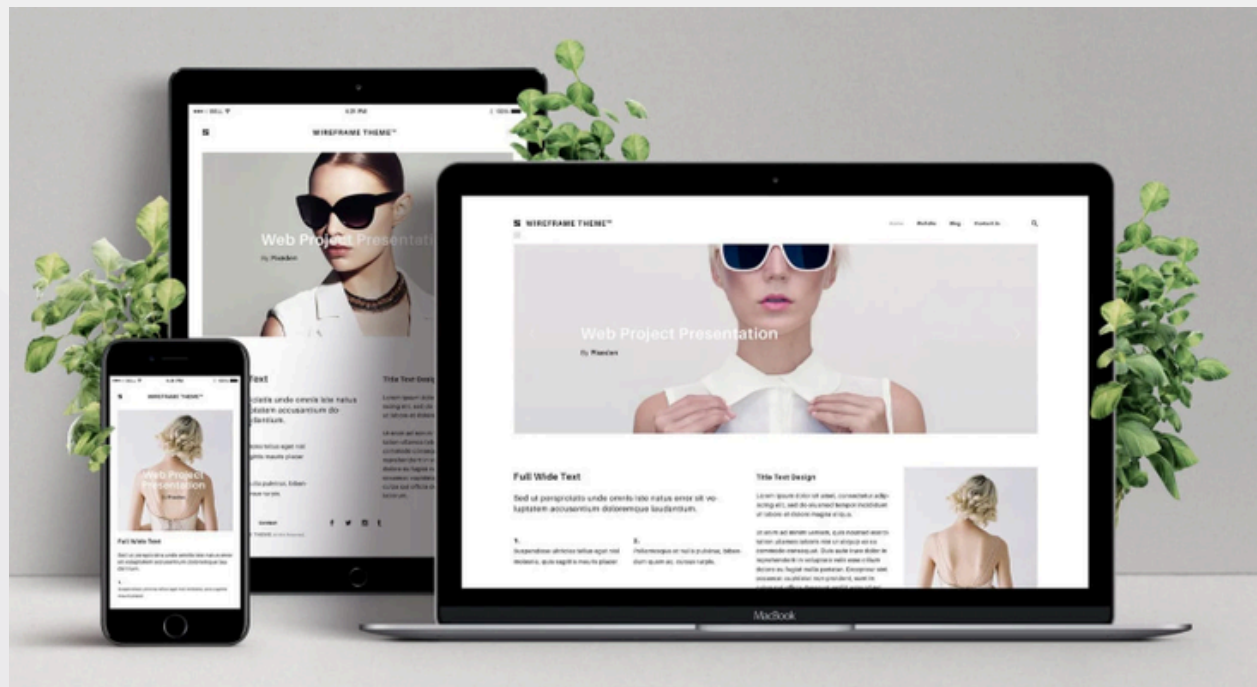


PHASE 3: VISUAL DESIGN (UI)

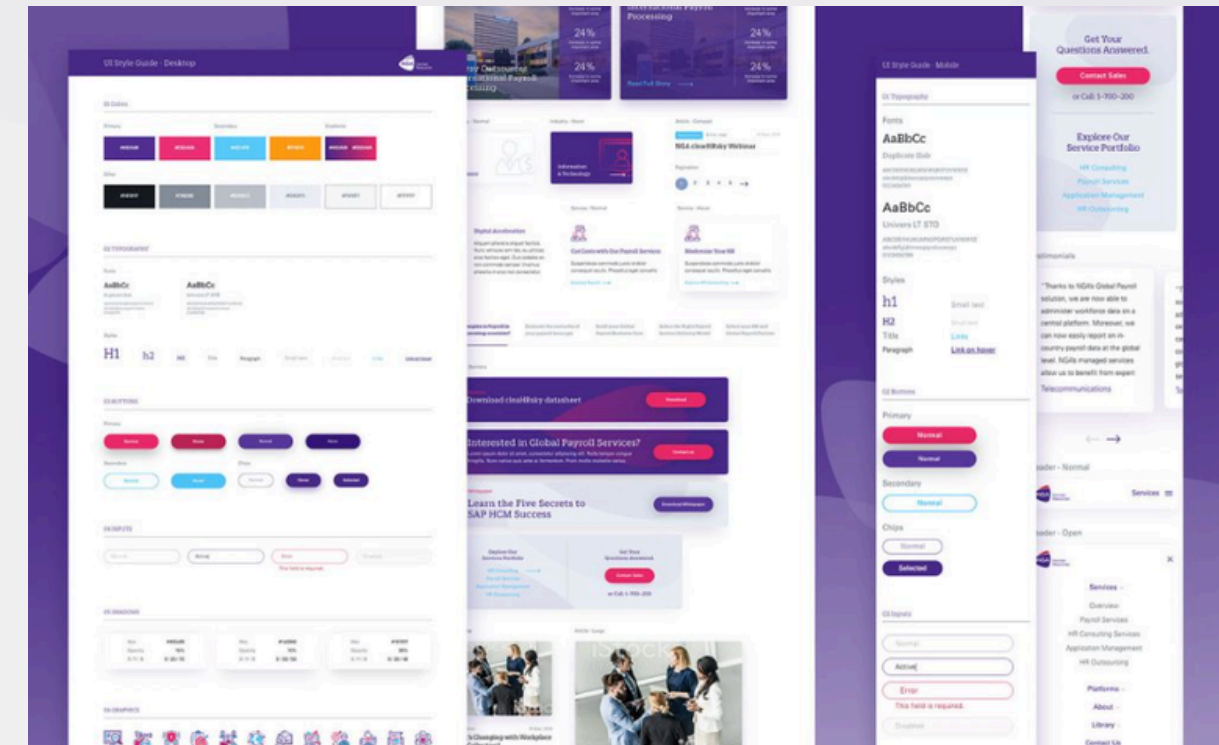
- **Objective:** Apply brand identity, color theory, and visual polish to wireframes.
- **Key Deliverables:**
 - High-fidelity design **mockups**.
 - **Responsive layouts** (desktop, tablet, mobile).
 - **Interactive prototypes**.
 - A **design system** or **component library**.

PHASE 3: VISUAL DESIGN (UI) (II)

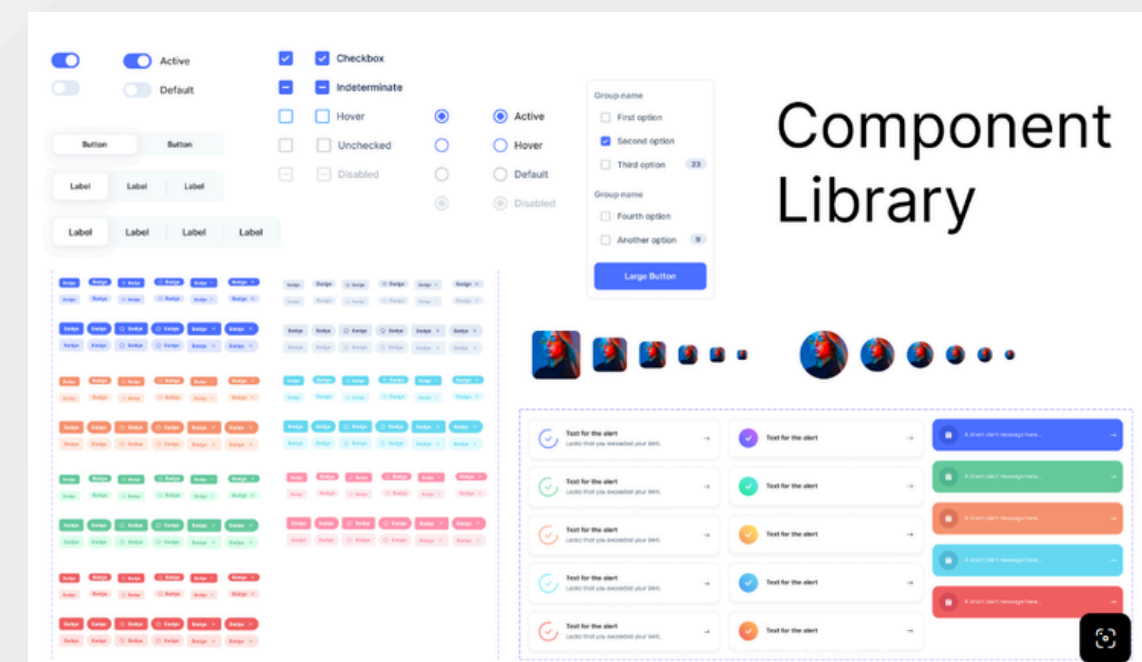
Responsive Layouts



Design system



Component library



Component Library

PHASE 4: DEVELOPMENT

- **Objective:** Turn visual designs into functional code.
- **Key Tasks:**
 - Front-end development (HTML, CSS, JavaScript/frameworks)
 - Back-end/API setup
 - Database architecture
 - CMS integration (e.g., WordPress, Webflow, custom headless setup).

PHASE 5: TESTING & QUALITY ASSURANCE (QA)

- **Objective:** Ensure functional integrity, performance, and accessibility across all platforms.
- **Key Tasks:**
 - Cross-browser and cross-device responsiveness.
 - Web accessibility compliance (WCAG 2.1 AA).
 - Load time optimization and technical SEO best practices.
 - Form submissions and interactive feature validation.

PHASE 6: LAUNCH & OPTIMIZATION

- **Objective:** Deploy the site to production and monitor performance.
- **Key Tasks:**
 - Domain migration
 - SSL setup, analytics tracking setup (e.g., Google Analytics)
 - post-launch bug fixes
 - continuous conversion rate optimization (CRO).

TASK 1

Define the project scope, business goals, and target audience for your **Intermodular Project**.

- **Key Tasks:**
 - **Analyze competitors:** at least look for two webpages from direct competition in the area and identify the sitemap. We will analyze their design system later.
 - **Create user personas**
 - Establish Key Performance Indicators (KPIs): for example, client satisfaction.

APPLICATIONS: ANALYSIS, DESIGN, AND PROTOTYPES

Why mockups and prototypes matter

- They define the structure and layout early
- They accelerate development
- They help clients visualize the final product
- They reduce misunderstandings and rework

PROTOTYPES. KEY ELEMENTS

- A prototype is a mockup or model of a design that helps visualize the final product.
- Prototypes allow designers to explore different layout options, test basic functionality, and evaluate usability.
- They save time, effort, and money, since it is much easier to make changes to an early design than to modify the final product.

WHY PROTOTYPES MATTER IN WEB DESIGN

- They allow multiple design iterations
- They help refine menus and interface elements
- They involve the client early in the process
- They enable functionality and usability testing
- They reduce cost and rework before coding begins

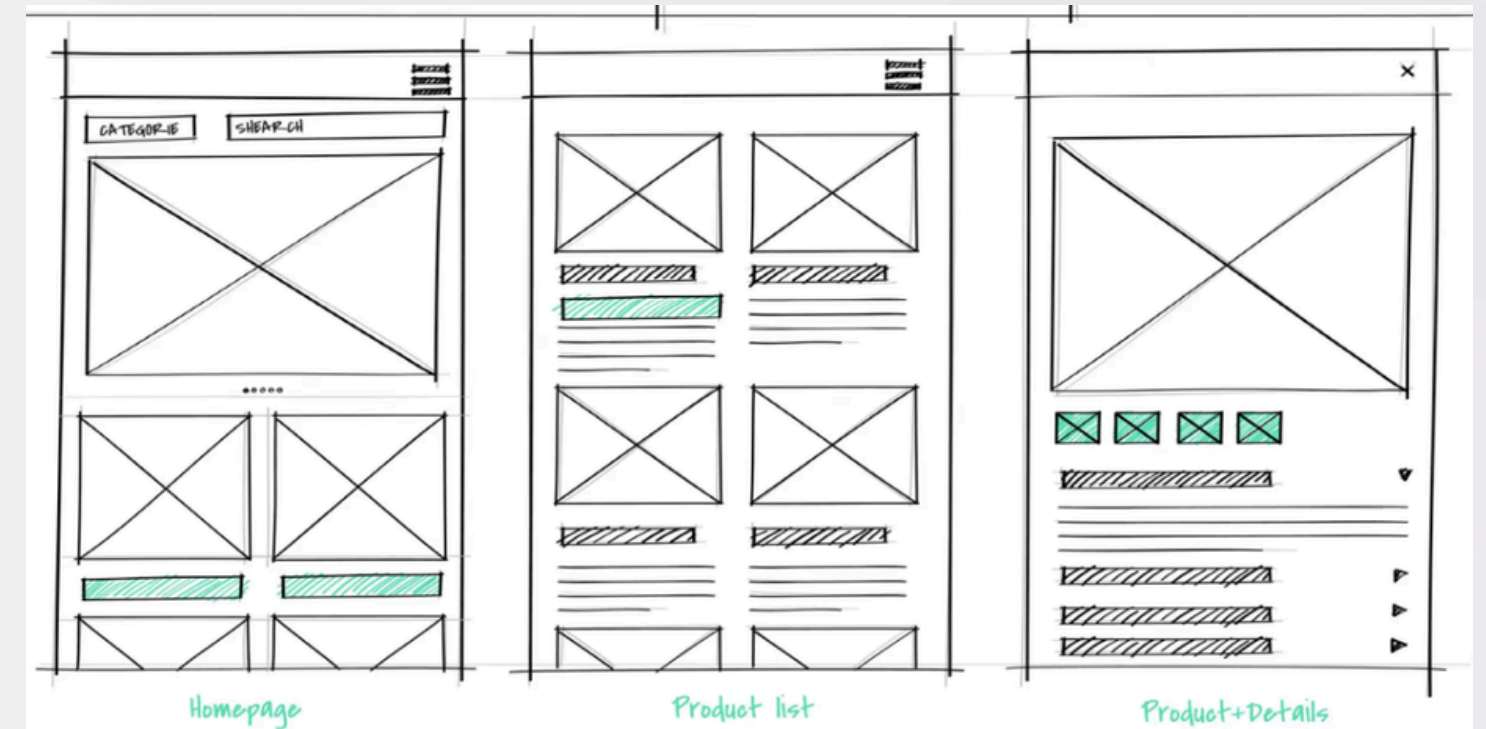
TYPES OF PROTOTYPES: SKETCHING

- Paper-based drawings of the whole website
- Includes processes and screen relationships
- Used in the early design phase
- Defines content hierarchy
- No visual design details yet



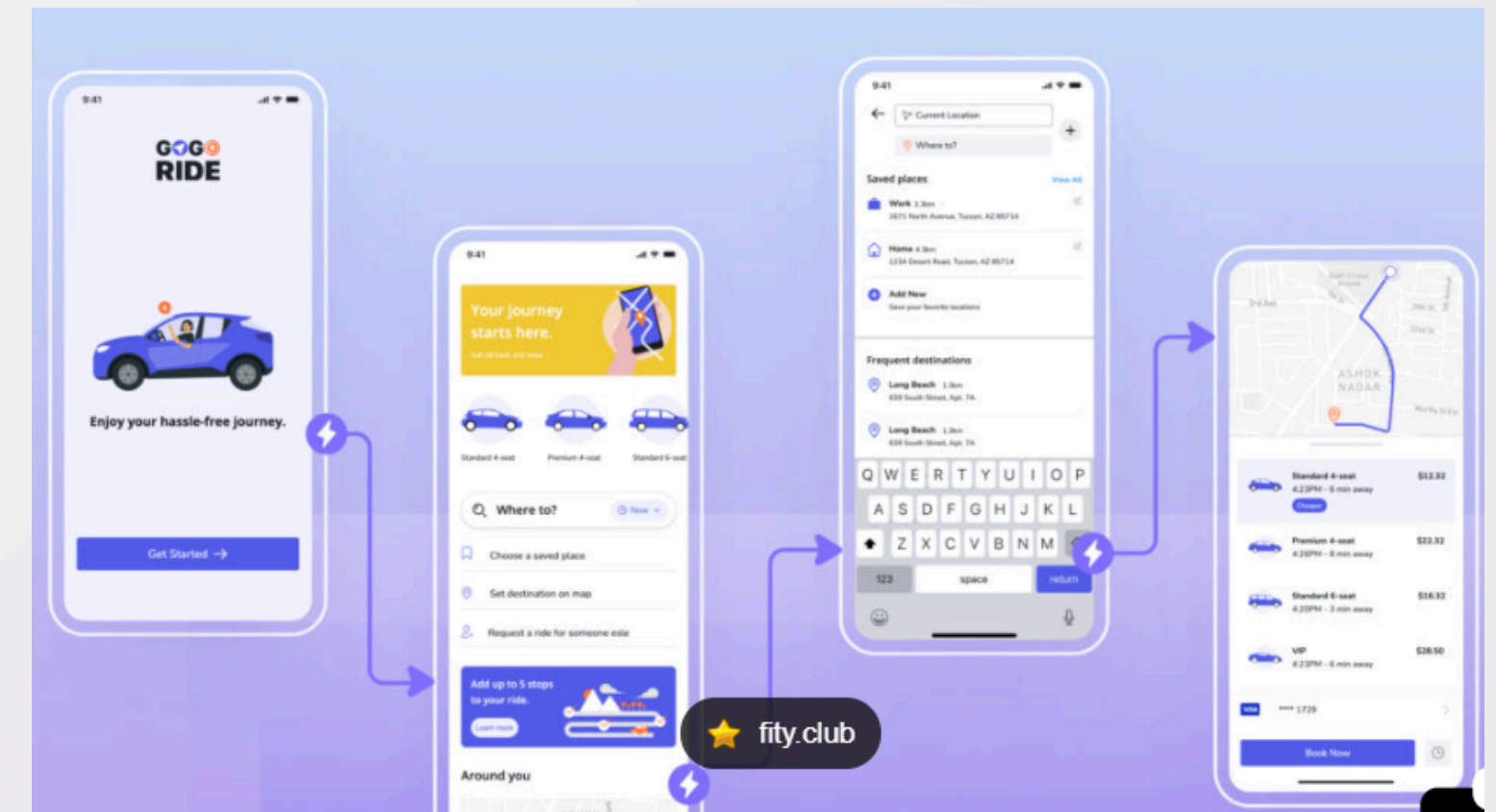
TYPES OF PROTOTYPES: WIREFRAMING

- More detailed screen drawings
- Includes content structure and **calls to action**
- Paper or digital format
- Produces a mock-up based on earlier sketches
- Helps validate design decisions with the client
- Main goal: content organization



TYPES OF PROTOTYPES: PROTOTYPING

- Digital design of screen interactions
- Final step before full development
- Evaluates design, content flow, and functionality
- Includes menus, forms, buttons, icons...
- Enables user testing
- Reduces development time and number of app versions



WIREFRAMING ORGANIZATION

The design of the prototype, in any of its phases, should be based mainly on the following aspects:

- Identifying the elements that belong to each page of the website.
- Distributing the number of graphical interface elements so that the screen is not overloaded, while still providing enough information and ensuring correct interaction.
- Organizing the hierarchy, order, and arrangement of elements.
- Ensuring an appropriate layout that efficiently uses the available space depending on the device.
- Selecting web design patterns to standardize the interface design.
- Evaluating technical aspects of usability and accessibility.

WIREFRAMING ORGANIZATION (II)

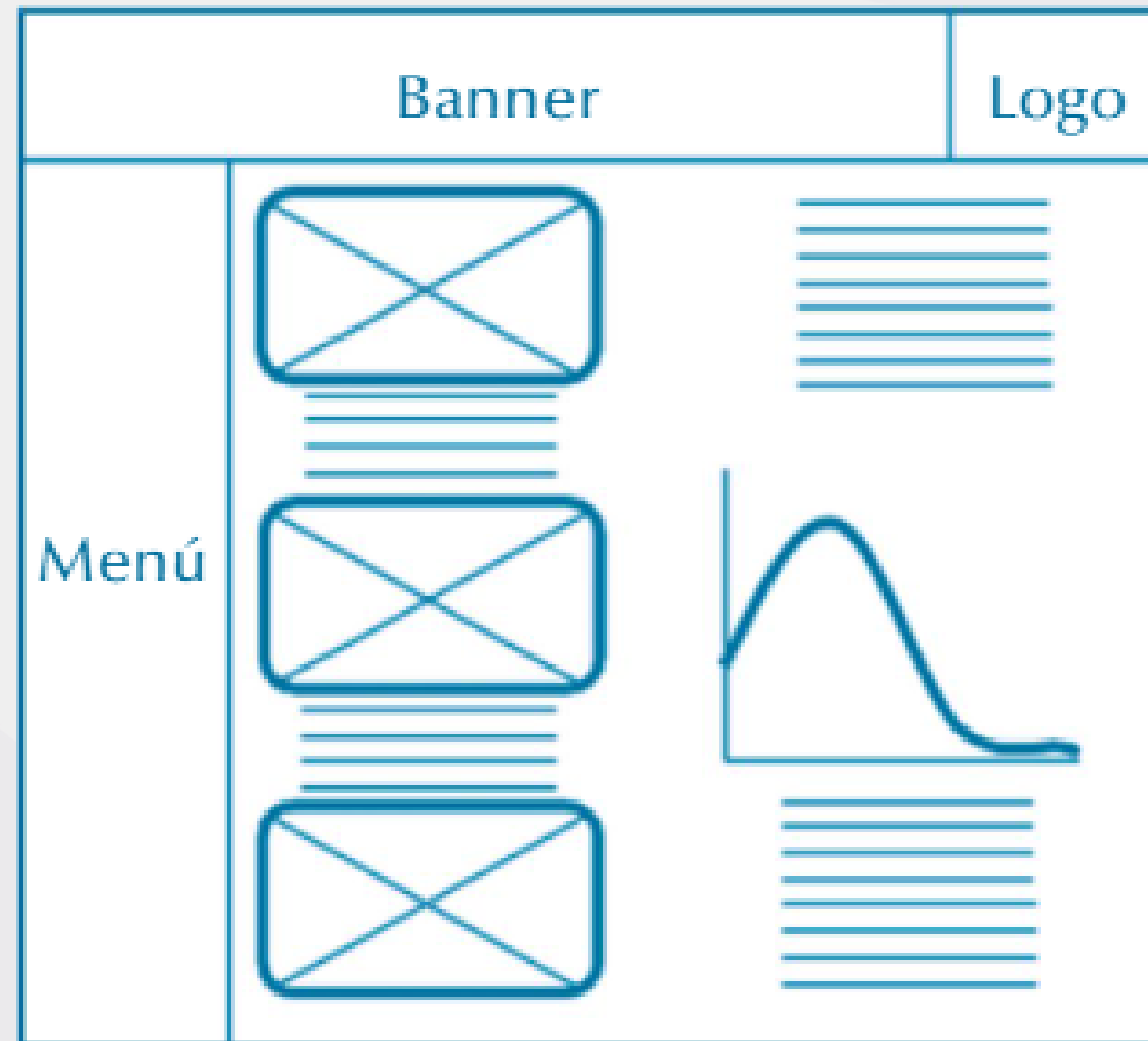
When discussing graphical user interfaces, two clearly differentiated areas can be identified. These are:

- a) **Wireframes**, or specific content blocks, which **varies from one page to another**.
- b) **Generic layouts**, or **common parts** that are reused across multiple pages of the website (header, footer...).

WIREFRAMES: INITIAL STRUCTURE OF A WEB INTERFACE

- Begin by creating a schematic representation of the main parts of the interface.
- Build a wireframe, a low-fidelity visual prototype that acts as the skeleton of the website or web application.
- Use wireframes to define the initial layout and distribution of content.
- Avoid including final design elements; wireframes are created in black and white or grayscale.
- Focus on structure, hierarchy, and placement—not aesthetics.

WIREFRAMES: INITIAL STRUCTURE OF A WEB INTERFACE (II)



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ADVANTAGES OF WIREFRAMES

- One of the main advantages of wireframes is that their creation focuses solely on content architecture, not on visual design.
- Although prototypes are the tools that allow testing content interaction, some applications enable web page prototyping, meaning you can click through links and share early interactive versions with clients.
- These interactive wireframes help validate structure and navigation before building the final product, reducing development time and minimizing the number of required versions.

TASK 2A

Create a sketching and wireframing to design the user interface for the inter-module project (including mobile devices).

Use pencil (or pen) to quickly sketch the different screens and user flows, including the home screen.

Experiment with different layouts and design elements

WEB APPLICATIONS FOR CREATING WIREFRAMES

- **Figma** – Collaborative, cloud-based tool for creating wireframes, mockups, and interactive prototypes.
- **Adobe XD** – Intuitive interface design tool with strong prototyping features.
- **Balsamiq** – Low-fidelity wireframing tool that mimics sketch-style layouts for quick ideation.
- **Sketch** – Popular macOS design tool with plugins for wireframing and prototyping.
- **MockFlow** – Web-based platform for planning interfaces and creating structured wireframes.
- **Wireframe.cc** – Minimalistic tool focused on fast, distraction-free wireframe creation.

FIGMA: A COLLABORATIVE INTERFACE DESIGN PLATFORM

- Cloud-based tool for designing **wireframes, mockups and interactive prototypes**.
- Enables **real-time collaboration**, allowing multiple users to work on the same file simultaneously.
- Provides an intuitive interface with **vector editing**, reusable components, and design libraries.
- Includes built-in **prototyping features** to simulate navigation and user flows.
- Works directly in the browser—no installation required.
- Ideal for team projects, classroom activities, and professional UI/UX workflows.

TASK 2B:

DIGITAL PROTOTYPING IN FIGMA

- Based on the paper prototype created in Task 2a, now move to the digital design phase.
- Use Figma to recreate the interface, maintaining the structure and layout defined in your sketches.
- Translate each screen into a digital format, applying consistent spacing, alignment, and component organization.
- Ensure that user flows designed on paper are fully represented in the digital version.
- Prepare the prototype for future interaction testing and refinement.

CONTENT MANAGEMENT SYSTEMS (CMS)

- In addition to prototyping tools, there are applications dedicated to **content management**, known as **CMS (Content Management Systems)**.
- These programs allow users to **create, edit, and publish web content** through a graphical interface **without requiring programming knowledge**.
- CMS platforms are important because they provide an **attractive and consistent design**, adding value that can make users prefer one website over another.
- The choice of a CMS depends largely on the **purpose and requirements of the project**.

TYPES OF CMS

- CMS platforms can be classified into **two main categories**:
 - **Proprietary software** – Requires a valid license to use.
 - **Open-source software** – Allows developers to access and modify the source code, making customization easier.
- Most well-known CMS platforms are built using **web technologies** such as **PHP/HTML** and rely on database management systems like **MySQL**.

TEMPLATES IN CMS

- Building a website with a CMS relies on predefined design elements known as **templates**.
- Templates can be **fully customized** by the user from scratch.
- Alternatively, CMS platforms offer **pre-designed sites**, which provide a ready-made structure where users only need to add their specific content.
- These templates simplify development, ensure visual consistency, and speed up the creation of functional websites.

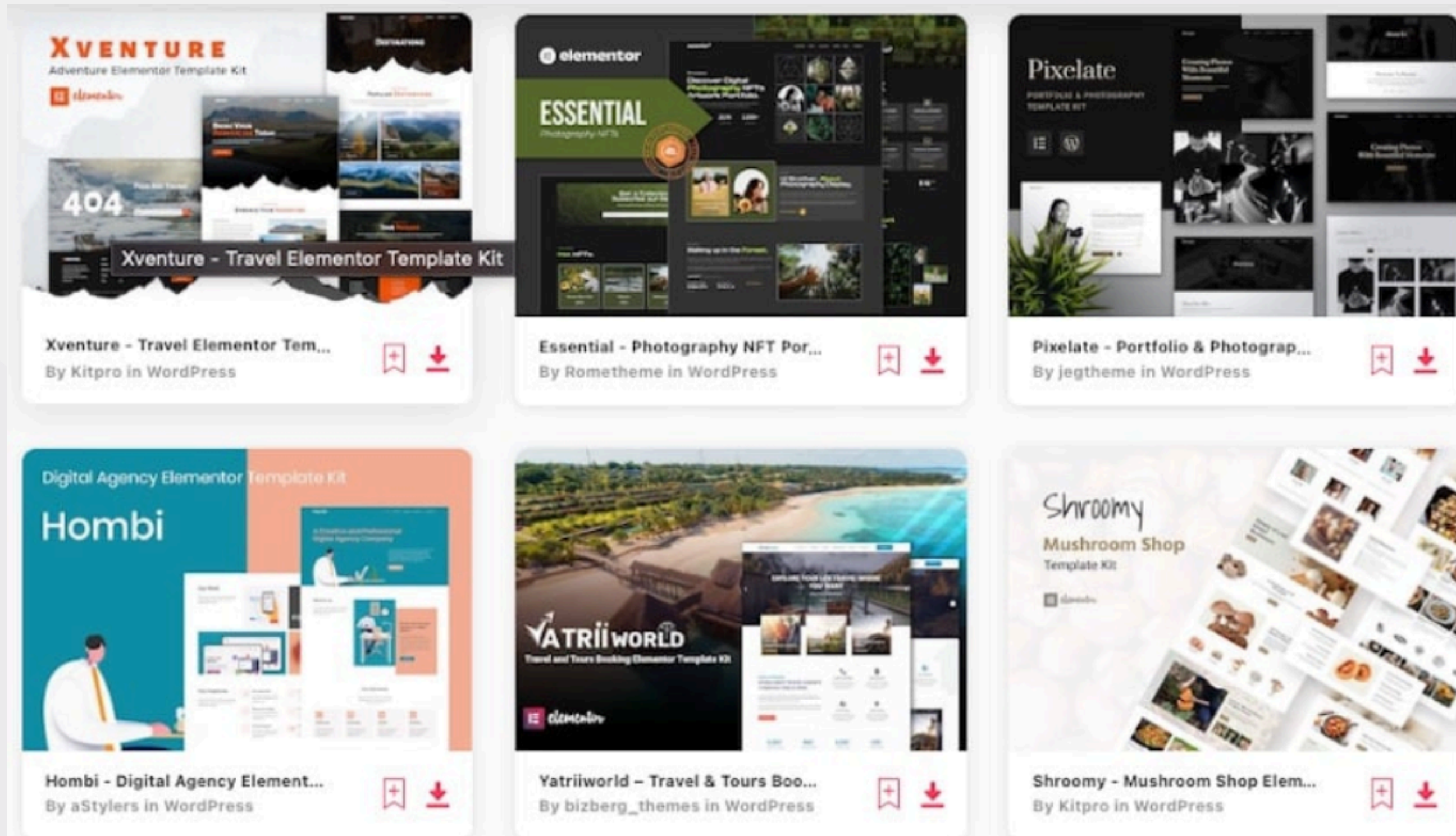
EXAMPLES OF CMS

- **WordPress** – The most widely used CMS; ideal for blogs, corporate sites, and small businesses.
- **Joomla** – Flexible and powerful, suitable for more complex websites and community platforms.
- **Drupal** – Highly customizable and secure; used for large-scale, enterprise-level projects.
- **Moodle** – A CMS specialized in e-learning; widely used in educational institutions.
- **Prestashop / Magento** – CMS platforms focused on e-commerce and online stores.
- **Ghost** – Modern, minimalist CMS designed for professional publishing and newsletters.

WORDPRESS

- WordPress is the world's most widely used CMS, powering more than 40% of all websites on the internet.
- Allows users to **create, edit and publish website** content through an intuitive dashboard—no advanced coding required.
- Offers thousands of **themes** to control the site's appearance and **plugins** to extend functionality (SEO, security, e-commerce, forms, etc.).
- Available in two versions:
 - **WordPress.com** – hosted, easier for beginners.
 - **WordPress.org** – self-hosted, fully customizable.
- Suitable for blogs, business sites, portfolios, e-commerce stores, educational platforms, and more.

WORDPRESS THEMES



RESOURCES

Unplash (Royalty Free) images: <https://unsplash.com/es>

Google Fonts: <https://fonts.google.com/>

ADDITIONAL READINGS

[Guidelines in Human Computer Interface \(HCI\)](#)

[Interaction in HCI: Principles, types, and examples](#)

[How to create wireframes as a UX Designer](#)

[Design Systems 101: What are they and why do they matter?](#)

[How to Create a Website Wireframe in Figma \(Beginner Tutorial\)](#)

[How to Wireframe in Figma in 2026](#)

ACRONIMS

HCI: Human-Computer Interaction

I/O or IO: Input/Output

SME: Small and medium-sized enterprises

UX: User Experience

UI: User Interface

¿ANY QUESTION?